

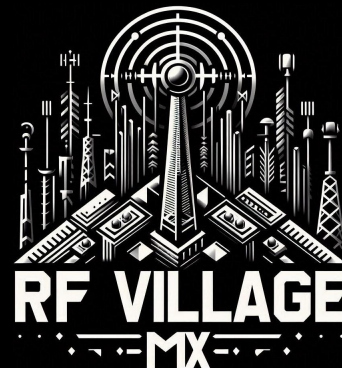


Lo que cuentan los satélites

Recibiendo telemetría
satelital por diversión

Manuel Rábade

manuel@rabade.net



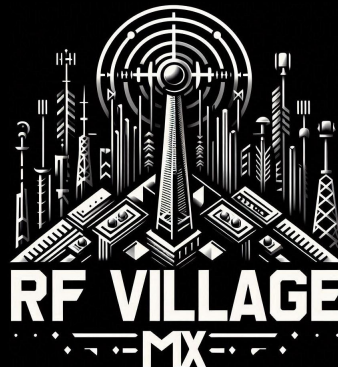


@manuelrabade

rabade.net



rf-village-mx.github.io



Introducción

- ¿Qué es telemetría?

Datos que un sistema remoto envía para monitoreo: posición, estado, sensores...

- ¿Para qué recibirla?

Aprender, experimentar y divertirse



En onda corta

— ¿Qué tal se escucha?

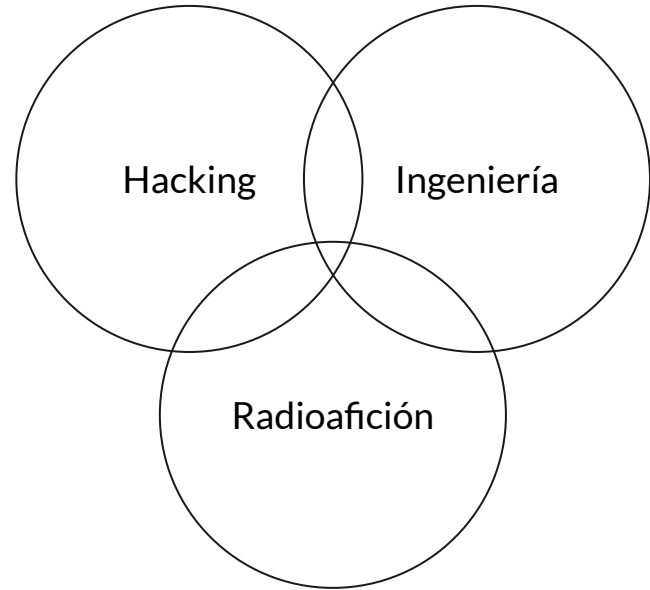
— ¡Bien!



Contenido

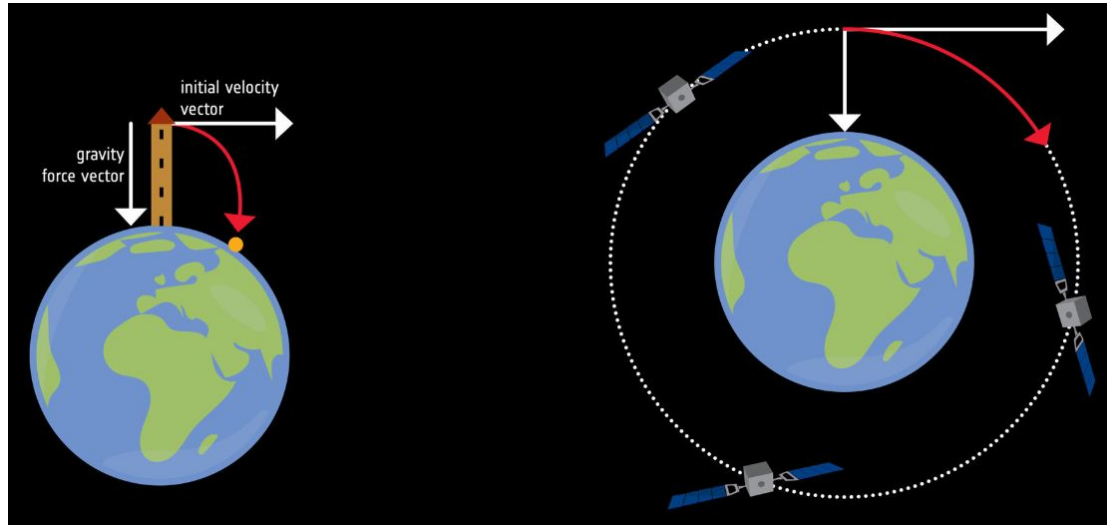
- Satélites
- Proyectos para recibir telemetría
- Construir una estación receptora
- Casos de estudio, uso y diversión

- Predicción de órbitas
- Comunidad en México



Satélites

¿Qué es un satélite?





Comparando satélites y aviones

Vehículo	Altitud	Velocidad	Peso	Periodo
Aviones comerciales	10 - 13 Km	900 km/h	70 - 400 t	-
CanSats	15 Km	-	0.05 - 0.1 Kg	-
Aviones militares	15 - 20 Km	2,000 km/h	10 - 40 t	-
Concorde	15 - 18 Km	2,180 km/h	78 t	-
Satélites en órbita baja (LEO)	200 - 2,000 Km	28,000 km/h (7.8 km/s)	100 - 2,000 Kg	90 - 120 min
Estación espacial internacional	400 - 420 Km		420 t	
CubeSats	350 - 700 Km		1 - 20 Kg	
Satélites GPS (MEO)	20,200 Km	14,000 km/h (3.9 km/s)	1 - 2 t	2 - 12 horas
Satélites en órbita geoestacionaria (GEO)	35,786 Km	11,000 km/h (3.1 km/s)	2 - 6.5 t	24 horas

Tipos de órbitas

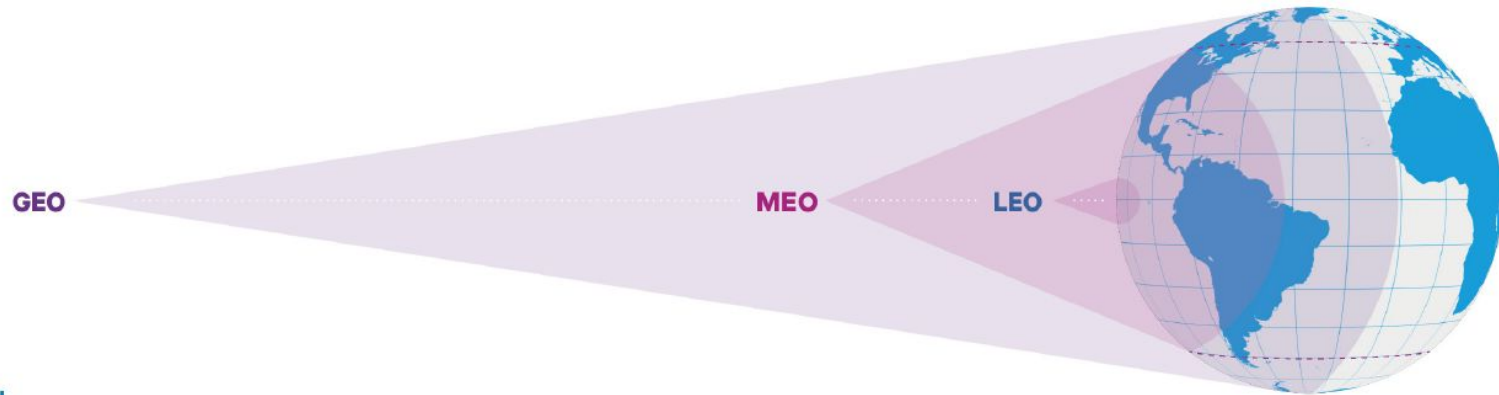
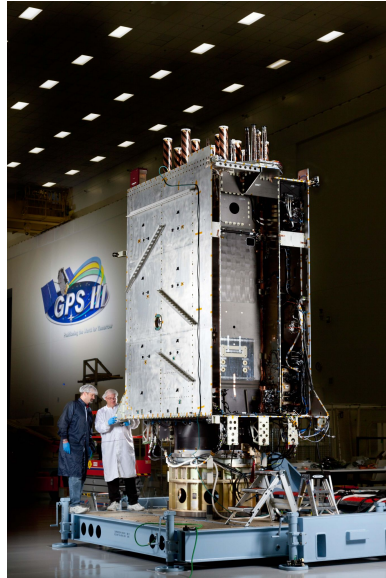
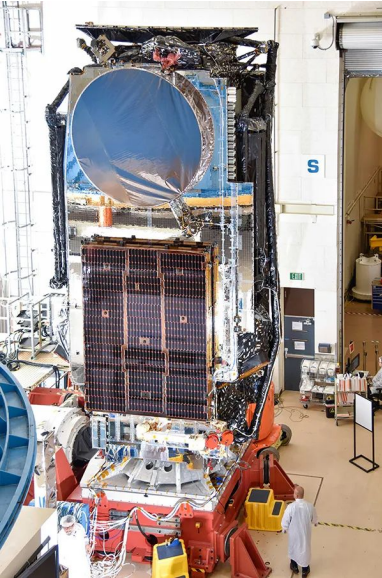
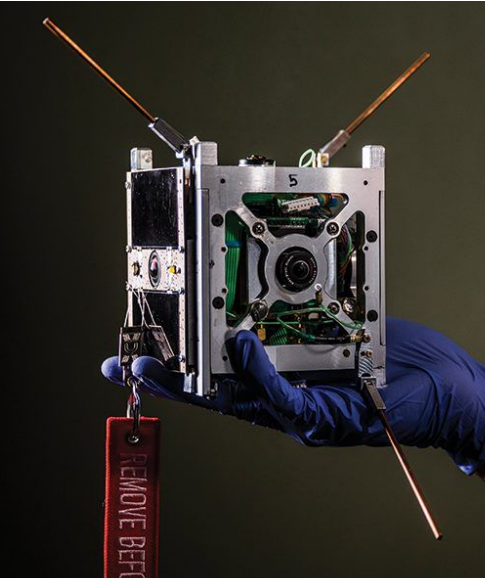


Figure 1: Schematic of orbital altitudes and coverage areas

Satélites en órbita geoestacionaria, media y baja

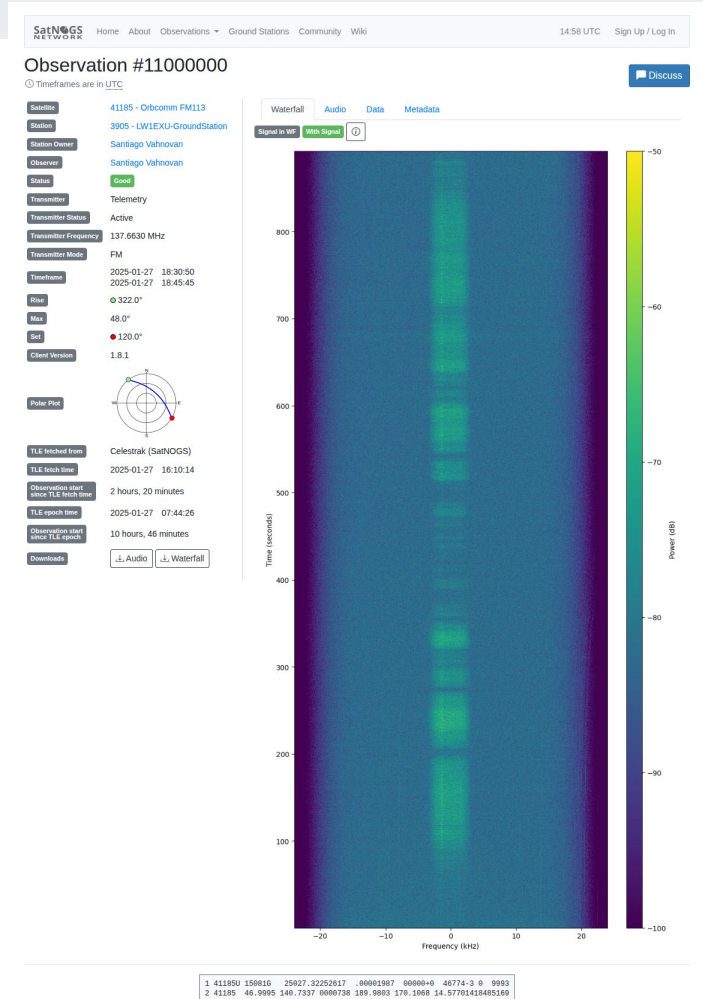
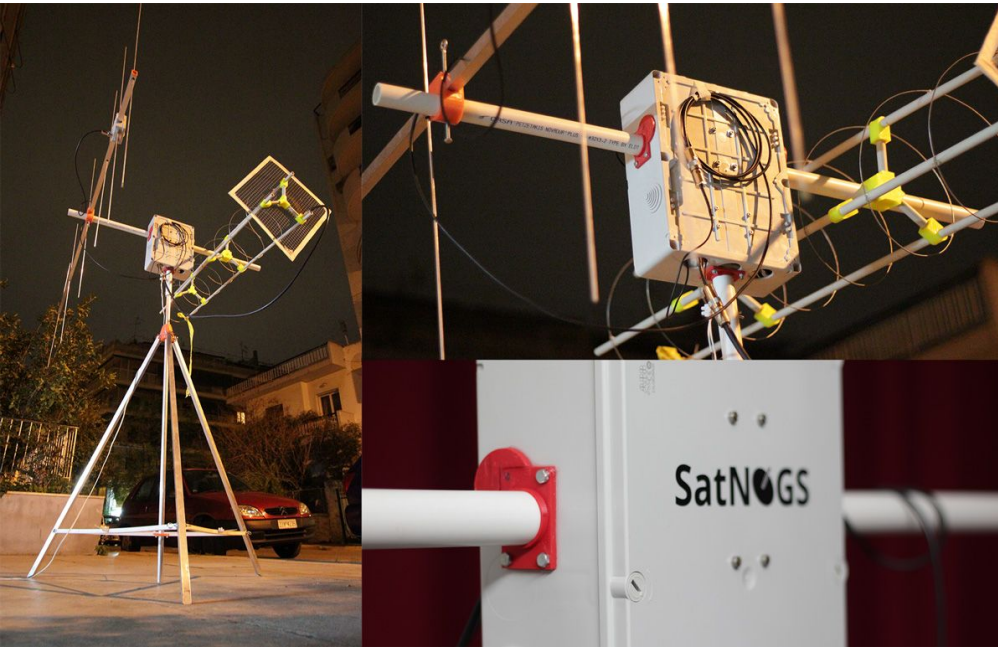


Satélites en órbita baja

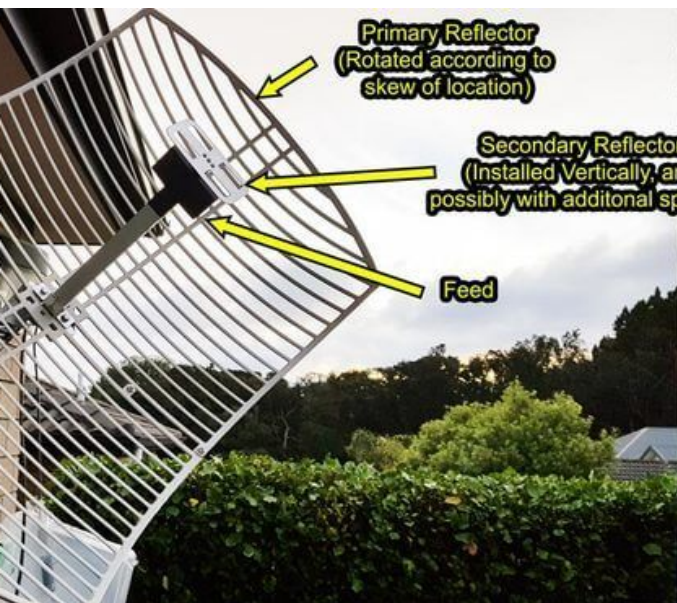


Proyectos para recibir telemetría

SatNOGS

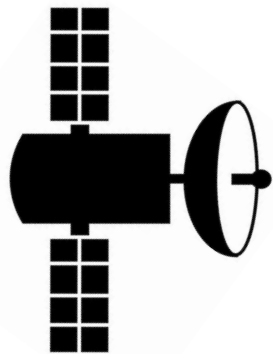


RTL-SDR.COM





TinyGS



LoRa



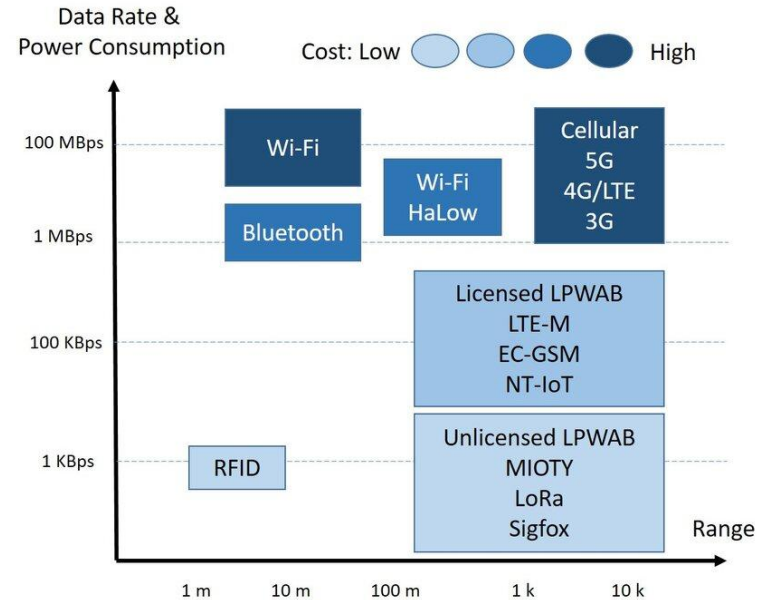
Internet



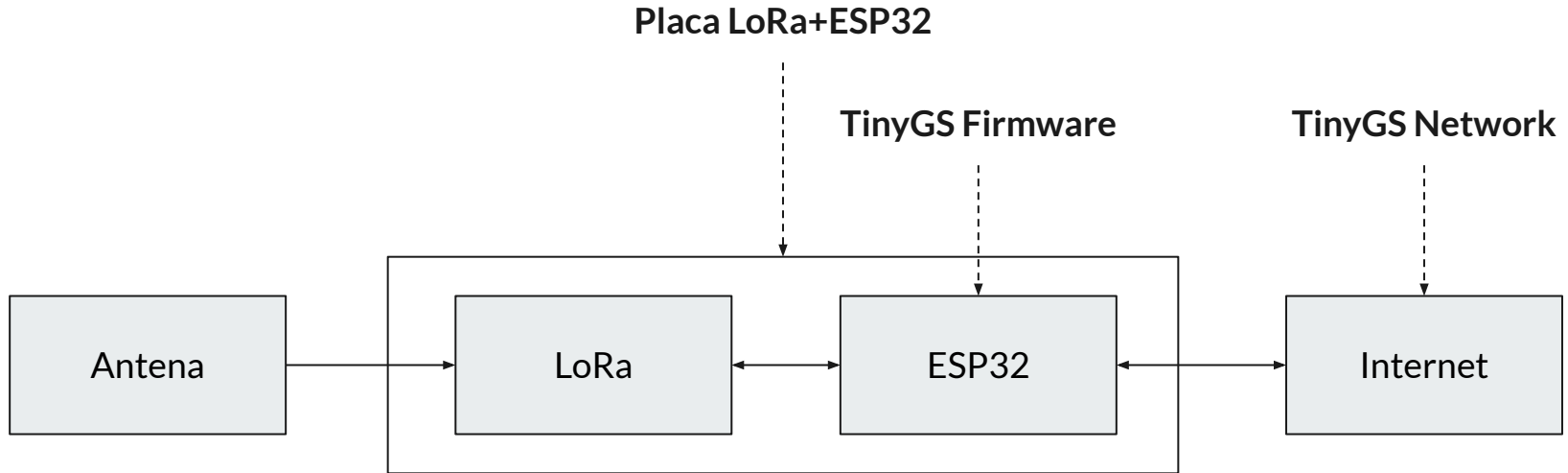
Construir una estación receptora

Modulación LoRa

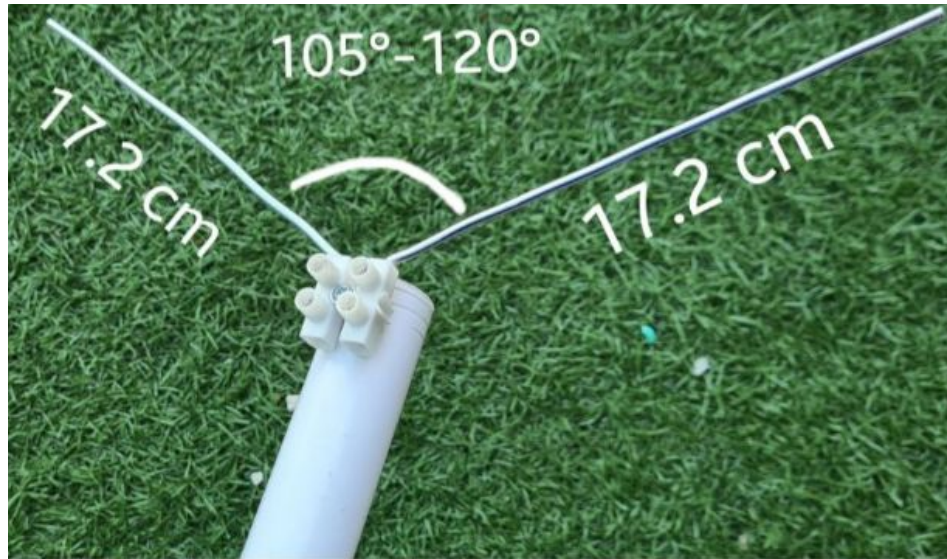
- LOnG RAnge
- Bitrate: 0.3 - 50 kbit/s
- Payload: 51 - 255 bytes
- Bandas que no requieren licencia ISM
 - 169 MHz
 - 433 MHz
 - 868/915 MHz
 - 2.4 GHz



Estacion receptora



Antenas DIY



Antenas comerciales



[Click to see full view](#)



UAYESOK 433Mhz Omni External Antenna for Remote Distance up to 1000 ft+, 3dbi Garage Door Electric Gate Opener Signal Booster, 433Mhz SMA Male GSM GPRS Aerial Device Receivers Alarm System

[Visit the UAYESOK Store](#)

4.0 ★★★★★ (35) | [Search this page](#)

Amazon's Choice

\$46⁹⁹

\$23.13 Shipping & Import Charges to Mexico [Details](#) ▼

Antenna	Radio
Brand	UAYESOK
Color	433Mhz SMA-Male 3M(9.8ft) RG58 Coaxial Cable
Impedance	50 Ohm
Maximum Range	9.8 Feet
Manufacturer	Quanzhou Hai Mao Wei Communication Co., Limited
Number of Channels	1
Number of Items	1



Zerone 433MHZ gsm GPRS SMA
Amplificador de señal de Antena de
claxon Macho Plug 3 Metros

★★★★★ (57)

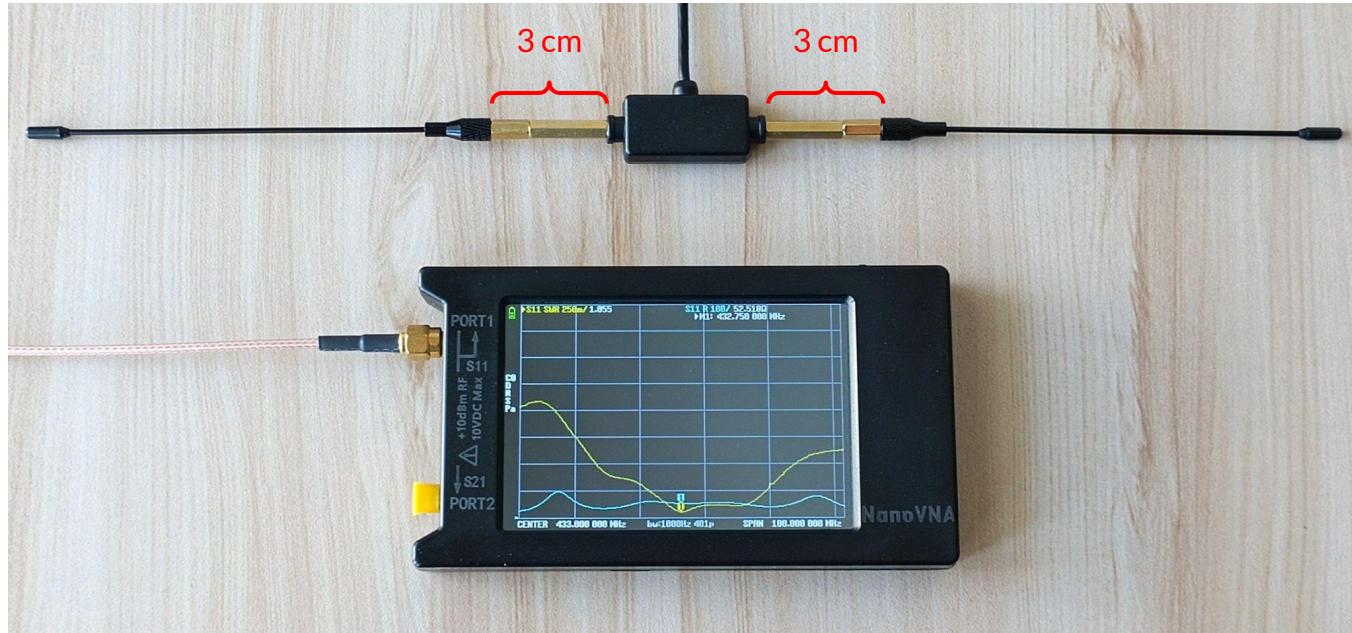
\$162⁰⁰

\$16.44 x 12 meses

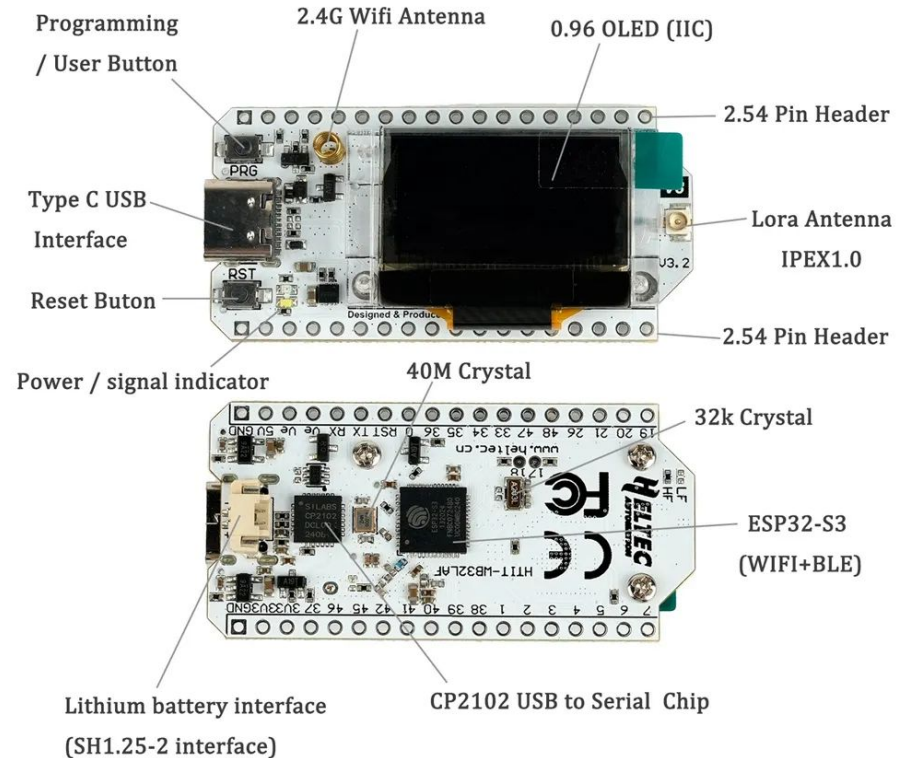
Entrega GRATIS entre el 11 - 20 de mar

Importación

Optimizando la antena



Placa LoRa+ESP32



Estación receptora



Lista de partes

Cantidad	Parte	Precio
1	Antena 433Mhz	\$ 169
1	LILYGO TTGO LoRa32 433 Mhz	\$ 599
2	Separador M3 - 20 mm	\$ 16
2	Separador M3 - 10 mm	\$ 10
	Total	\$ 794

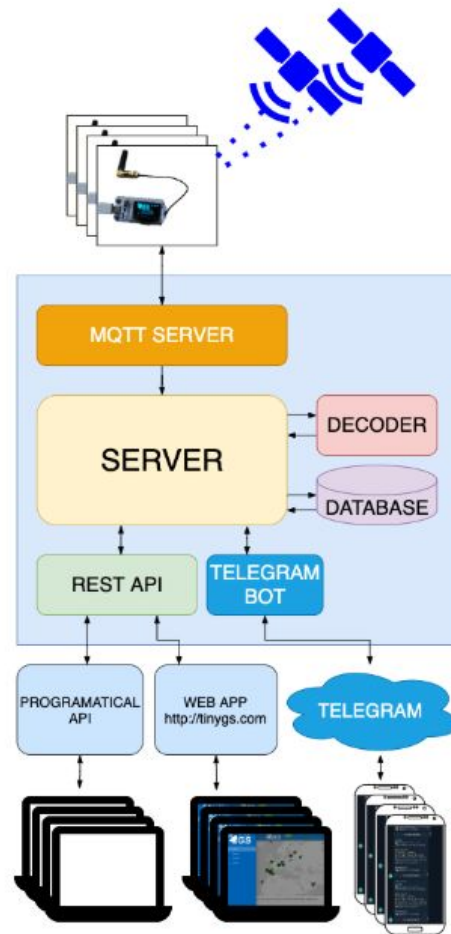


<https://gist.github.com/manuel-rabade/3e20432e063050936ab2534ee6deee56>

TinyGS



<https://tinygs.com>



tinyGS Community
6 926 members, 128 online

General 07:32
Hernedy Arrieta: 😊

Antenna Building 21:07
Lua: What kind of antenna would you rec...

Technical problems 14:30
Juan Pablo: 📶 Test you a rabbit ear, it's...

Beta firmware testing 13:49
Nick: The 868/915 MHz LoRa excels in I...

statistics 06:15
tinyGS: 📊 GaoFen 27/February/2025 📦 ...

Space situational awaren... 05:26
tinyGS: Found 📶 Tianqi-36 TLE 📶 TIAN...

Events:tinyGS Thu
Juan Pablo: 📷 Album, Does to its succ...

Learning Resources Wed
Germán: 📖 Understanding S Param...

TinyGS (share your setup) Tue
Carlo: TinyGS and MeshCom on the roof.

!! 🚀 New satellites Feb 23
Stefan/OE6ISP: Finally it's the CG

! 📢 Announcements Jan 29
G4lile0: Everything is back to normal op...

🛒 Where to buy Jan 27
wind: Anyone is using RAK19003?

Balloons Jan 24

General
In tinyGS Community

Pinned Message
This video might be interesting...

Juan Pablo
Stefan/OE6ISP
Maybe in autotune mode and nothing to tune so far ...
Tnks 10:55

Stefan/OE6ISP
Juan Pablo
Tnks
Can you see any sats in range on the map? 10:56

Antonio
Hola a todos desde Murcia, España / Hello everyone from Murcia, Spain
👍 🇳🇵 12:01

Today

Hernedy Arrieta
Hola Hernedy Arrieta desde Venezuela, Saludos. 07:00
Hello Hernedy Arrieta from Venezuela, Greetings. (YV1GAG) 07:01
What is the procedure to obtain the credentials and activate my station? 07:02

Stefan/OE6ISP
Hernedy Arrieta
What is the procedure to obtain the credentials and activate my station?
open @tinygs_personal_bot and enter /mqtt 07:13

😊 07:32

Message



<https://t.me/joinchat/DmYSElZahiJGwHX6jCzB3Q>



Welcome to the TinyGS web installer!

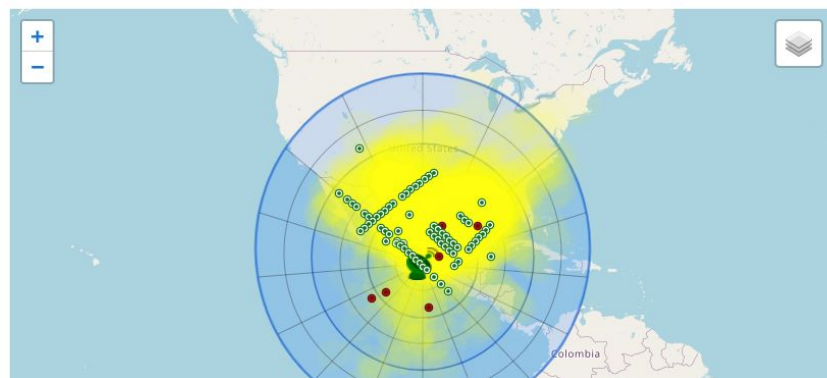
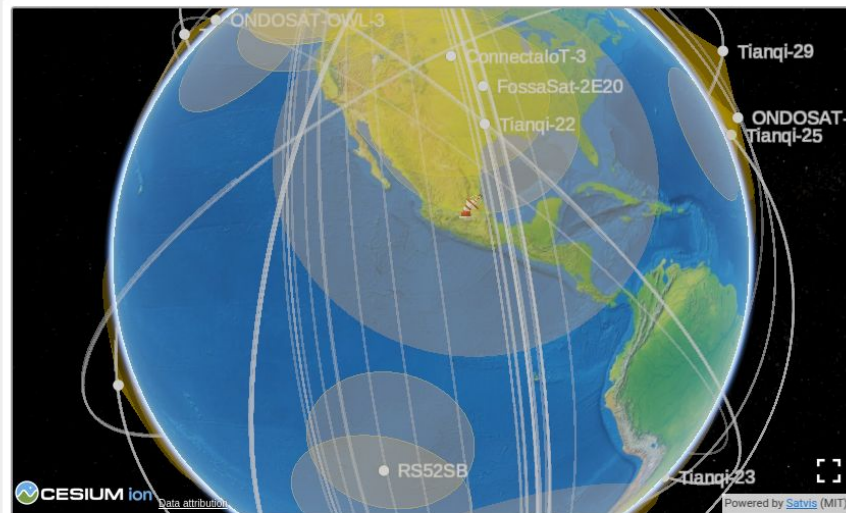
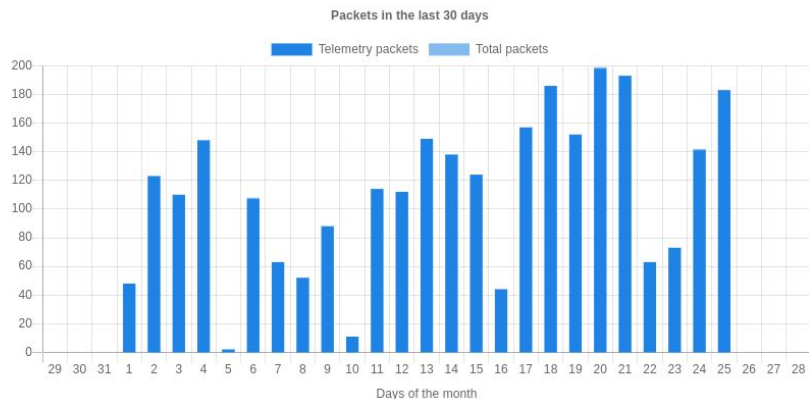
1. Plug in your ESP to a USB port. We will install TinyGS 2403242 to it.
2. Hit "Install" and select the correct COM port. [No device found?](#)
3. MQTT credentials (First join the group [here](#))
Then open a private chat with [@tinygs_personal_bot](#) and ask /mqtt
4. First time board boot, connect to the wifi AP "My TinyGS" to configure it.
5. Get TinyGS installed and connected in less than 5 minutes!

Install

2403242



Status	Listening	Version	Creation date
● Online	FossaSat-2E20	2403242	3 months ago
Last seen	Last Packet	Position (Lat, Long)	QTH Locator
a minute ago	2 days ago	19.385, -99.145	EK09kj
Elevation	Auto tuning	Test mode	Confirmed packets
2231.00 m	433	OFF	5023
Telemetry packets	Type of antenna	Band	Record distance
4996	Dipole	400 - 500 MHz	2506.8 Km





Feb 26, 2025 04:31:59 (3 days ago)

Frequency Error

-4020.24Hz

CRC Error

Received by

42 Stations

Mode

LoRa@436.703

Power

7000mW

Distance

564Km

Elevation

56.27°

RSSI

-119.75 dBm

SNR

-10.75 dB

Predicted Doppler

2418.86Hz



Feb 26, 2025 04:30:58 (3 days ago)

Frequency Error

-9909.04Hz

CRC Error

Received by

45 Stations

Mode

LoRa@436.703

Power

7000mW

Distance

799Km

Elevation

33.80°

RSSI

-118.25 dBm

SNR

-9.25 dB

Predicted Doppler

7842.07Hz



Feb 26, 2025 01:04:22 (3 days ago)

Frequency Error

4725.93Hz

CRC Error

Received by

1 Stations

Mode

LoRa@400.265

Power

9000mW

Distance

1204Km

Elevation

39.14°

RSSI

-127.25 dBm

SNR

-14.25 dB

Predicted Doppler

-5460.45Hz



Feb 26, 2025 01:03:07 (3 days ago)

Frequency Error

1005.58Hz

CRC Error

Received by

4 Stations

Mode

LoRa@400.265

Power

9000mW

Distance

981Km

Elevation

54.43°

RSSI

-127.5 dBm

SNR

-13.5 dB

Predicted Doppler

-2138.65Hz

Norby

Received on: November 19, 2024 4:42 AM

LoRa 436.703 Mhz SF: 10 CR: 5 BW: 250 kHz

Sat in Umbra  Eclipse Depth: 11.21°

Theoretical coverage 4853 km

 7000mW  21°C

 8238mV  1896mW  21°C

 0mW  13011mAh  -2482mW

 Board PMM: 12°C PAM: 12°C PDM: 9°C

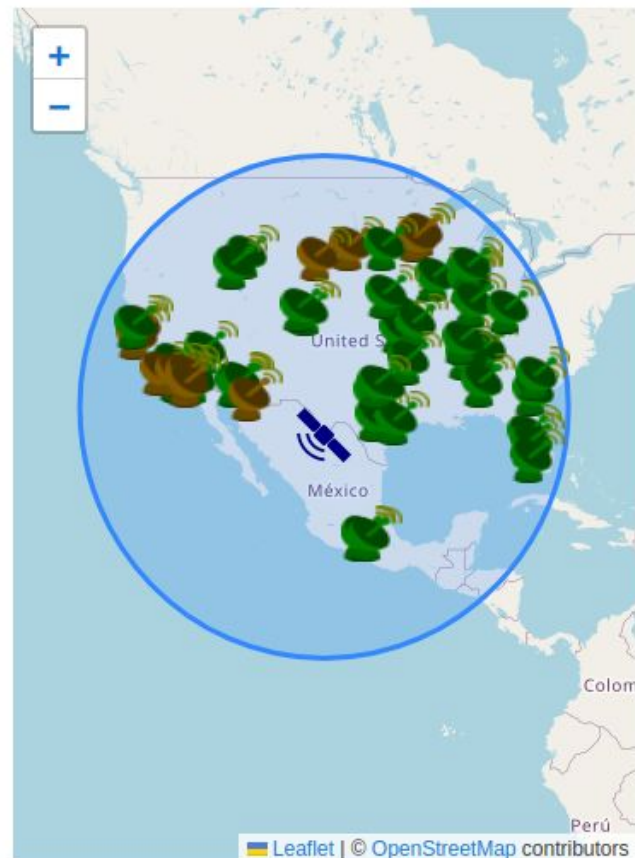
 Solar Array X-: -5°C X+: -6°C

 : 3013.42344

Hexadecimal view

[Download](#)

	0	1	2	3	4	5	6	7	8	01	2	3	4	5	6	7	8
0000	8E	FF	FF	FF	FF	0A	06	01	C9								
0009	34	84	00	00	00	00	F1	0F	00	4.....							
0012	00	68	A5	5C	8A	CF	2E	42	52	.h.\...BR							
001B	4B	20	4D	57	20	56	45	52	3A	K MW VER:							
0024	30	35	61	5F	30	31	00	00	00	05a_01...							



Casos de uso, estudio y diversión



Casos de uso, estudio y diversión

1. Recibir telemetría
2. Mejorar tu estación
3. Ingeniería inversa
4. Retransmitir información
5. Diagnóstico de satélites
6. Descubrir satélites

Norby

Received on: November 19, 2024 4:42 AM

LoRa 436.703 Mhz SF: 10 CR: 5 BW: 250 kHz

Sat in Umbra 🌑 Eclipse Depth: 11.21°

Theoretical coverage 4853 km

🏠 7000mW 🌡️ 21°C

🔋 8238mV 📶 1896mW 🌡️ 21°C

☀️ 0mW 📶 13011mAh 🌡️ -2482mW

🌡️ Board PMM: 12°C PAM: 12°C PDM: 9°C

🌡️ Solar Array X-: -5°C X+: -6°C

📶: 3013.42344

Hexadecimal view

[Download](#)

```
0 1 2 3 4 5 6 7 8 012345678
0000 8E FF FF FF FF 0A 06 01 C9 .....
0009 34 84 00 00 00 00 F1 0F 00 4.....
0012 00 68 A5 5C 8A CF 2E 42 52 .h.\...BR
001B 4B 20 4D 57 20 56 45 52 3A K MW VER:
0024 30 35 61 5F 30 31 00 00 00 05a_01...
```

4. Electrical power system

The NORBY electrical power system is designed to supply power to satellite subsystems. The functionality of the EPS includes the distribution of incoming and stored energy between NORBY subsystems according to a given algorithm, as well as monitoring the current status of system parameters. The EPS comprises four function blocks (figure 3):

- the power generation block;
- the power storage block;
- the electricity conversion block;
- the monitoring and control block.

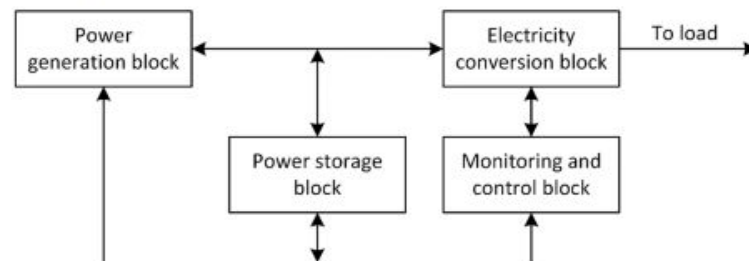


Figure 3. Functional diagram of the NORBY electrical power system.

The power generation block is a functional unit that converts solar energy into electrical energy and then transfers it to the other EPS blocks.

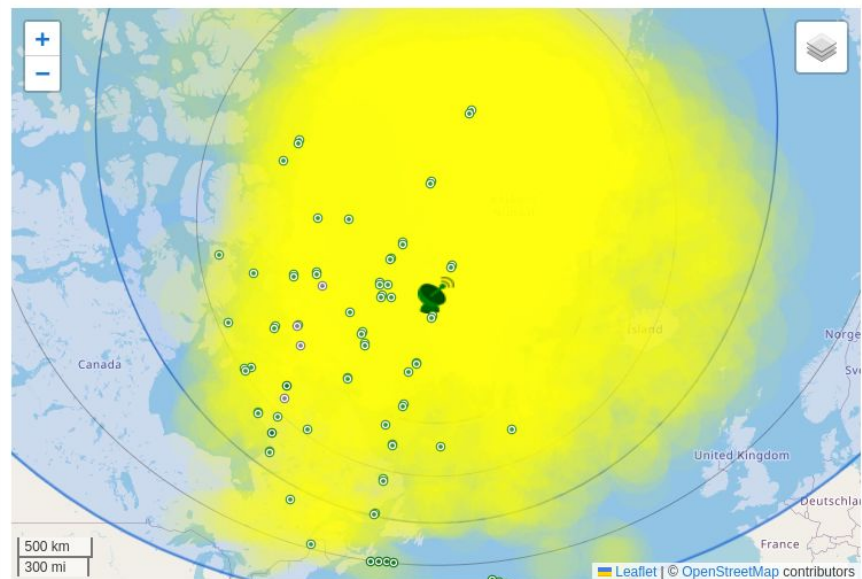
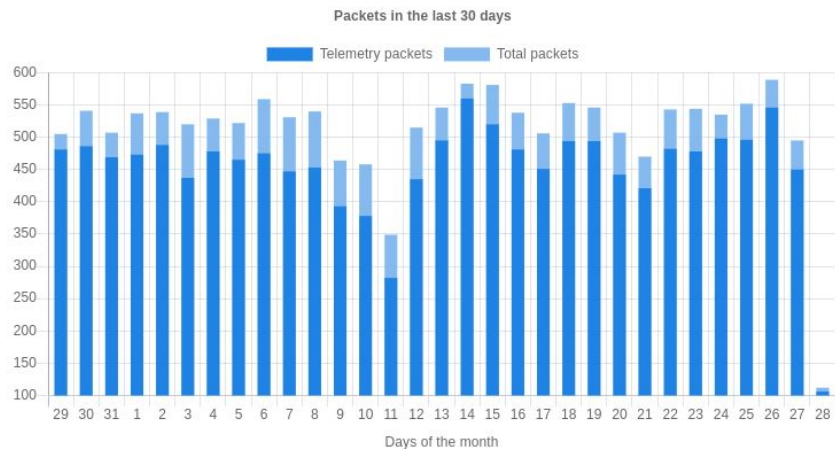
NORBY has six solar panels. Four of them are rigidly fixed to the outer sides of the satellite structure. Two of the panels are fold-out. Between 6 and 12 photovoltaic (PV) cells are located on each solar panel. Solar panels on the sunny side of the orbit generate electricity proportional to the power of incident light, with cell efficiency reaching 28% and a maximum output power of 1.25 W per PV cell. Each PV cell is connected to its own DC/DC converter, which ensures that maximum power is withdrawn from the cell. Each converter is daisy-chained to another of the same converters to produce a total output voltage in the range of 5 V to 8.3 V. At the output of each such pair is a diode to protect the entire solar panel in case one of the panel elements fails. The maximum power of the

https://tinygs.com/station/OX3HI_1@1762931706

Status	Listening	Version	Creation date
● Online	Surv-251	2403241	4 years ago
Last seen	Last Packet	Position (Lat, Long)	QTH Locator
a few seconds ago	18 minutes ago	67.011, -50.73	GP47pa
Elevation	Auto tuning	Test mode	Confirmed packets
37.00 m	433	OFF	305582
Telemetry packets	Type of antenna	Band	Record distance
300334	Eggbeater	401 - 479 MHz	3482.5 Km

Description

Antenna: EB-432 EGGBEATER Preamplifier: SSB Super-Amp SP-70 Receive: LILYGO® TTGO LoRa32 V1.1 433MHZ LoRa ESP-32 OLED 0.96 Inch





Galileo

@g4lile0.bsky.social

+ Seguir

My Sunday Sudoku is reverse-engineering the new  Starlink telemetry format. 😊



26 de enero de 2025, 5:54  Todos pueden responder

HOD-HOD-1A

Received on: January 1, 2025 1:19 PM
LoRa 436.25 Mhz SF: 9 CR: 6 BW: 250 kHz
Sat in Sun 🌞 Eclipse Depth: -32.68°
Theoretical coverage 4813 km

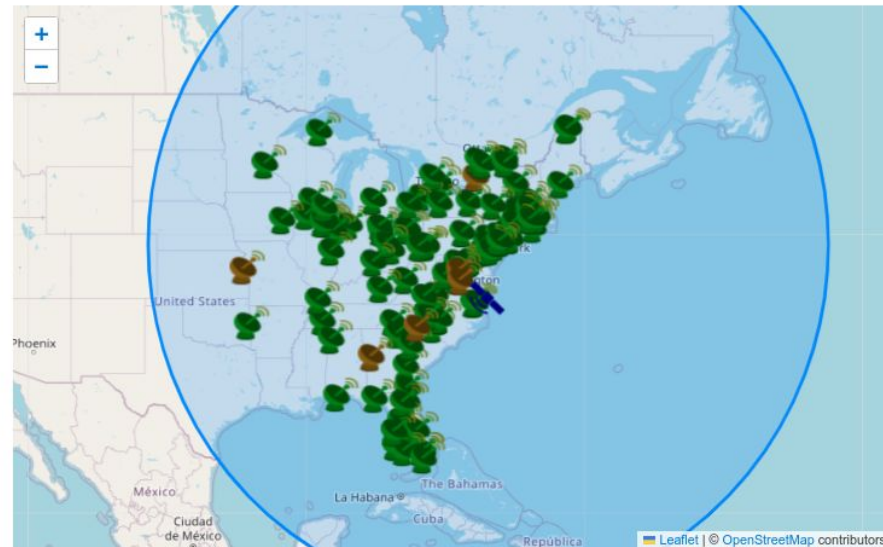
Hexadecimal view

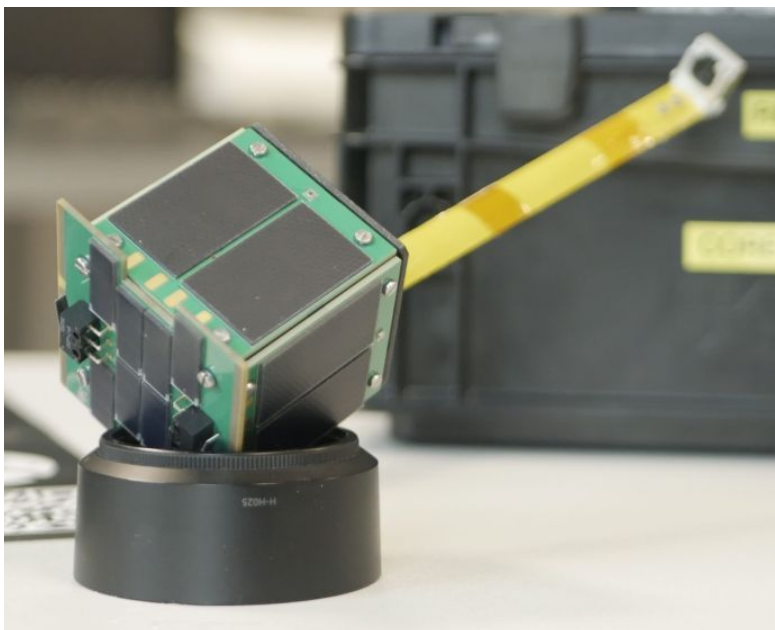
[Download](#)

0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	0123456789ABCDEF
0000	54	49	4E	59	47	53	20	4B	34	4B	44	52	20	48	65	6C TINYGS K4KDR He'l
0010	6C	6F	20	66	72	6F	6D	20	46	4D	31	37	21	20		lo from FM17!

Raw parsed view

▶ { ... }

[copy](#)



Filip Tomczyk / SQ3TLE

Kubuś

Quick summary about HYPE as for now: the mission began nominally a...

Technical update on that:

We reproduced the issue about 16 hours after the anomaly on the engineering model in our lab, using the last valid telemetry data from TinyGS ❤️. Lab testing showed us that at temperatures below 5°C, the SX1268's PLL failed to lock within the timeframe specified by our software. As a result, the radio entered an invalid state, reported failure to initialize to OBC and rest of the satellite continued working in safe mode.

Thankfully, our "dead man switch" mechanism rebooted the satellite after 24 hours of no valid, cryptographically signed commands from the ground. On the second attempt, 48 hours after the initial failed reboot, the satellite successfully restarted at a higher temperature.

At the moment, we're operating with the awareness that HYPE cannot be rebooted in low temperatures without facing a potential wait of one to few days. However, a software update to fix this issue is now scheduled for upload within the next two weeks.

Fun fact:

During preflight thermal tests, we verified continuous operation from -40°C to +60°C with the pass condition being no restarts, so this bug managed to slip through 🤔



17:38

<https://bsky.app/profile/tinygs.bsky.social/post/3lct7qx64fs2s>



TinyGS

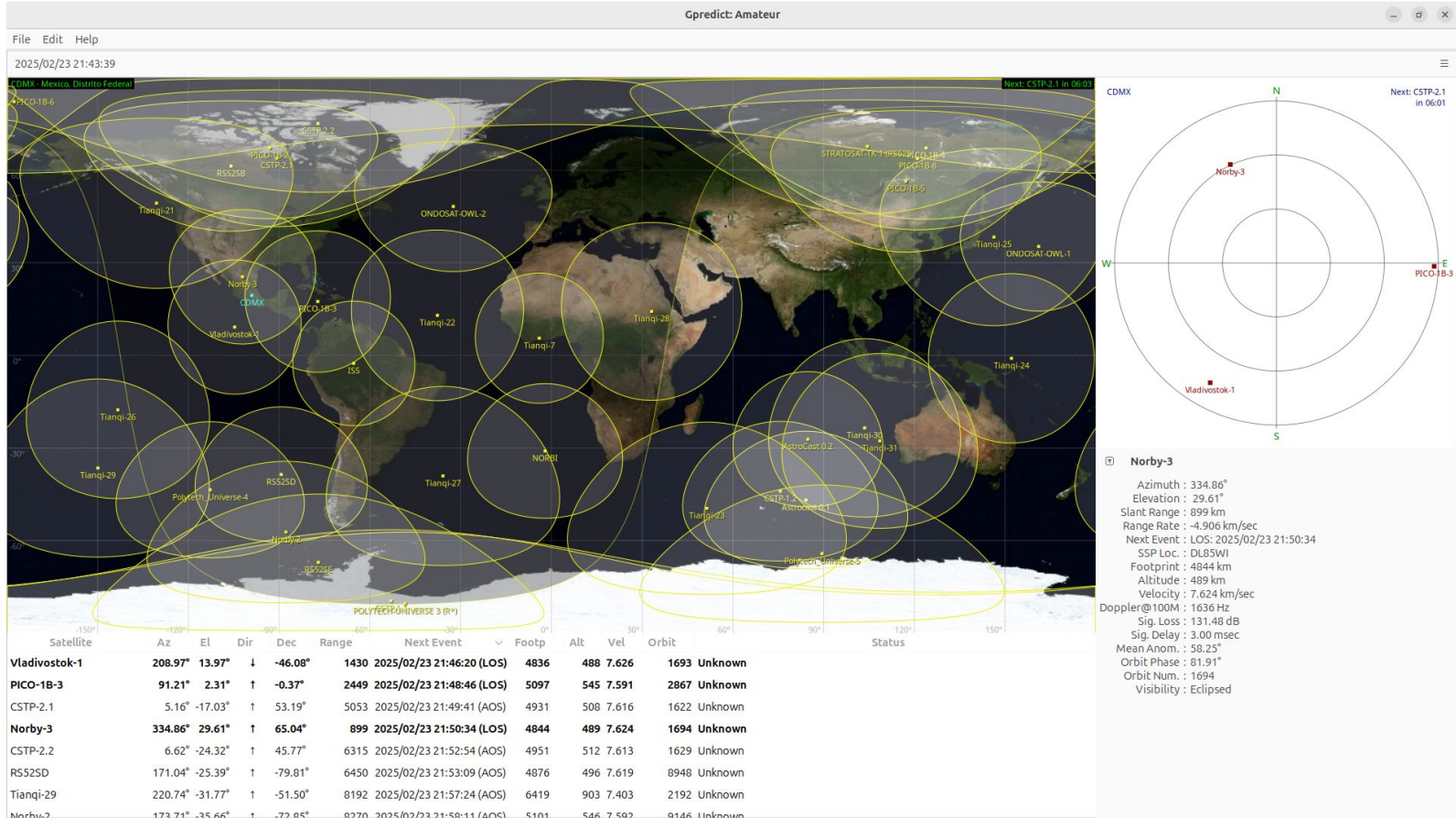
@tinygs.bsky.social

Found 🗑️: HOD-HOD-1A ➡️ fits TLE 2024-199AK. Our
[@tinygs.bsky.social](https://bsky.app/profile/tinygs.bsky.social) community of over 1600 ground stations is
enhancing Space Situational Awareness by matching TLEs! [#ESA](#)
[#satellite](#)

Predicción de órbitas

<https://oz9aec.dk/gpredict/>

https://api.tinygs.com/v1/tinygs_supported.txt



Upcoming passes for NORBI

AOS	LOS	Duration	Max El	AOS Az	LOS Az
2024/11/20 14:45:51	2024/11/20 14:54:07	00:08:15	8.08°	119.39°	25.42°
2024/11/20 16:17:44	2024/11/20 16:28:43	00:10:59	30.92°	186.17°	334.86°
2024/11/21 03:57:14	2024/11/21 04:08:25	00:11:10	47.83°	19.00°	181.76°
2024/11/21 15:56:42	2024/11/21 16:08:06	00:11:24	70.75°	172.00°	345.75°
2024/11/22 03:36:44	2024/11/22 03:47:12	00:10:28	22.09°	30.45°	167.07°
2024/11/22 05:10:46	2024/11/22 05:19:49	00:09:02	11.22°	340.43°	232.95°
2024/11/22 15:35:57	2024/11/22 15:47:13	00:11:15	49.50°	158.02°	356.27°
2024/11/23 03:16:33	2024/11/23 03:25:33	00:08:59	10.60°	43.87°	150.57°
2024/11/23 04:49:19	2024/11/23 04:59:47	00:10:27	22.82°	352.74°	216.66°
2024/11/23 15:15:30	2024/11/23 15:26:03	00:10:33	23.09°	143.46°	7.15°
2024/11/23 16:49:43	2024/11/23 16:58:50	00:09:06	10.85°	209.06°	316.45°
2024/11/24 04:28:09	2024/11/24 04:39:19	00:11:10	49.26°	3.71°	201.98°
2024/11/24 14:55:26	2024/11/24 15:04:33	00:09:07	11.39°	127.27°	19.41°
2024/11/24 16:28:02	2024/11/24 16:38:36	00:10:34	22.46°	192.68°	329.74°
2024/11/25 04:07:11	2024/11/25 04:18:30	00:11:18	70.08°	14.34°	187.87°
2024/11/25 16:06:44	2024/11/25 16:18:00	00:11:15	48.55°	178.04°	341.14°
2024/11/26 03:46:28	2024/11/26 03:57:20	00:10:52	30.23°	25.41°	173.51°

Print

Save

Close

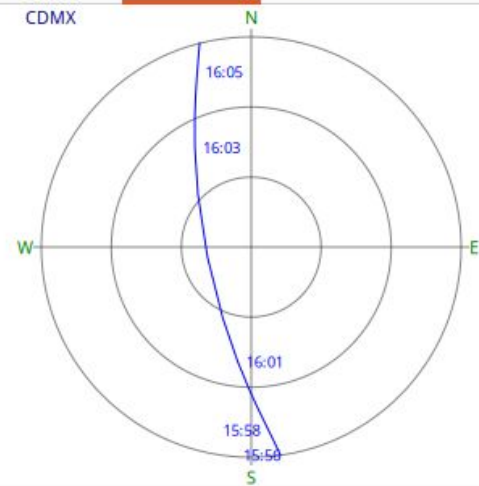
Pass details for NORBI (orbit 22841)

Data

Polar

Az/El

CDMX



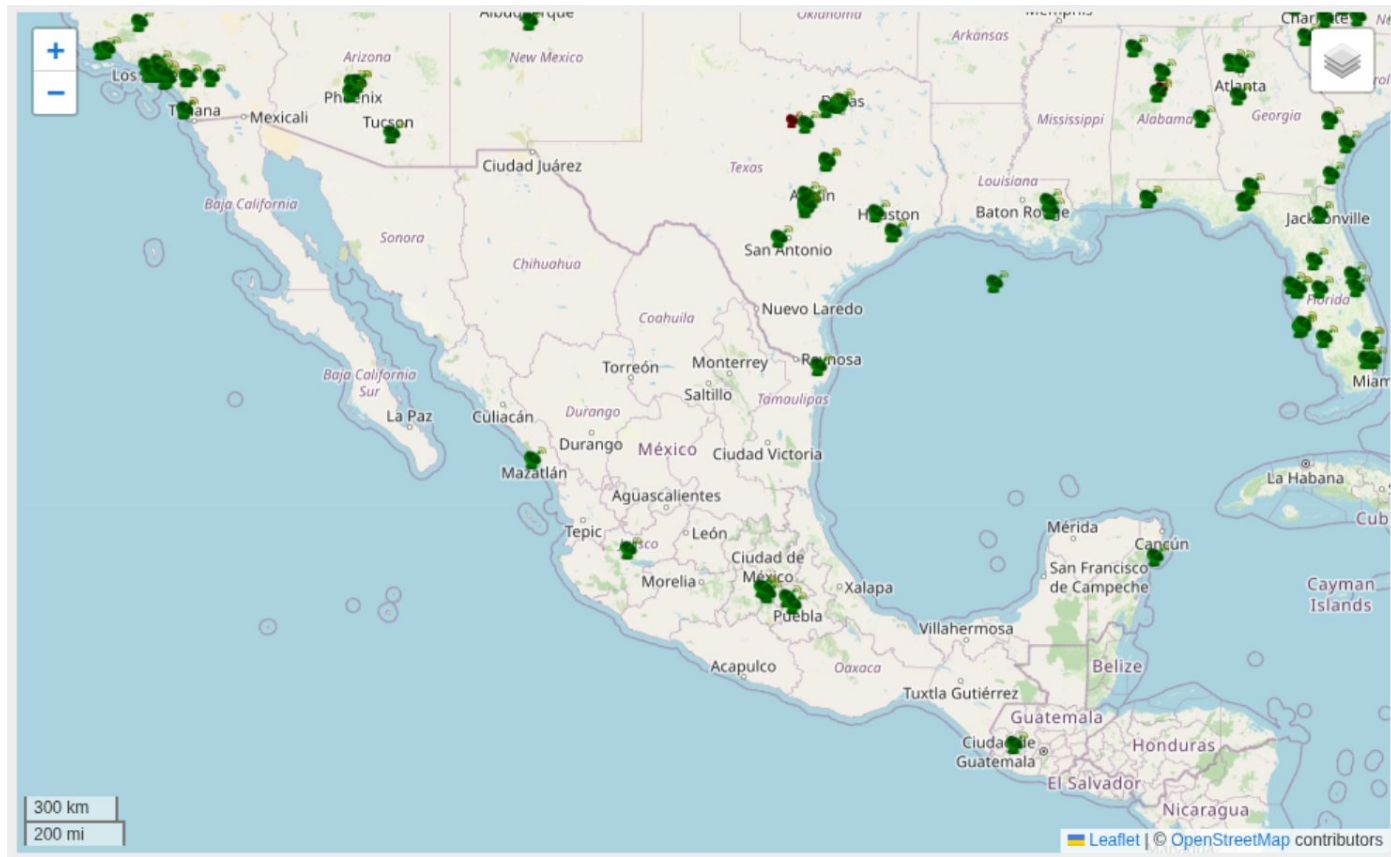
Print

Save

Close

Sig. Loss : 153.47 dB

Comunidad en México





Reunión virtual junio 2025



@manuelrabade

manuel@rabade.net



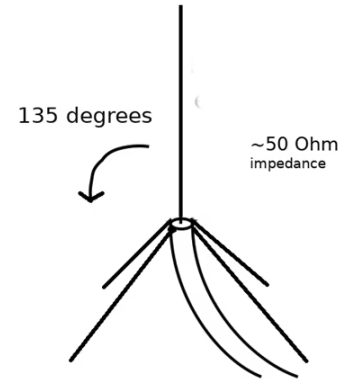
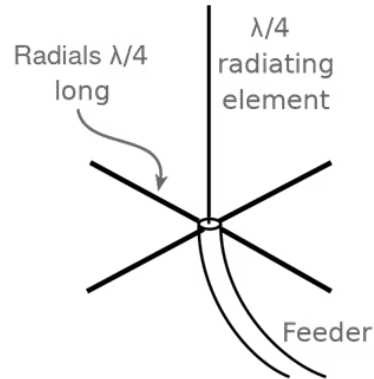
Estación KPR

Antena plano de tierra

$$f = 433 \text{ Mhz}$$

$$\begin{aligned}\lambda &= c / f \\ &= (3 \times 10^8 \text{ m/s}) / 433 \text{ Mhz} \\ &= 0.692 \text{ m}\end{aligned}$$

$$\lambda/4 = 17.3 \text{ cm}$$



<https://community.element14.com/challenges-projects/project14/rf/b/blog/posts/building-a-poor-man-s-quarter-wave-433mhz-antenna-antenna-s-construction>

Lista de partes

Cantidad	Parte	Precio
1 pcs	HELTEC LoRa32 433 Mhz	\$ 490
1 pcs	Cable RG174 SMA Macho-Macho 1 metro	\$ 45
5 pcs	Barras de cobre 2.5 mm	\$ 60
1 pcs	Conector UHF hembra para panel	\$ 60
1 pcs	Adaptador SMA hembra a UHF macho	\$ 135
4 pcs	Terminales de ojillo	\$ 24
4 pcs	Tornillos con tuerca	\$ 20
	Total	\$ 835



<https://gist.github.com/manuel-rabade/828e2fe1408905fdd48a35a22b0b2e72>



Configurar NanoVNA

DISPLAY > TRACE > TRACE 0 > BACK
DISPLAY > FORMAT > SWR
DISPLAY > CHANNEL > S11

DISPLAY > TRACE > TRACE 1 > BACK
DISPLAY > FORMAT > RESISTANCE
DISPLAY > CHANNEL > S11

DISPLAY > TRACE > TRACE 2 > BACK
DISPLAY > FORMAT > SMITH
DISPLAY > CHANNEL > S11

DISPLAY > TRACE > TRACE 3 > BACK
DISPLAY > FORMAT > LOGMAG
DISPLAY > CHANNEL > S12



Calibrar NanoVNA

STIMULUS > CENTER -> 433 Mhz
STIMULUS > SPAN -> 200 Mhz (100 Mhz)

CALIBRATE > RESET
CALIBRATE > CALIBRATE > OPEN
CALIBRATE > CALIBRATE > SHORT
CALIBRATE > CALIBRATE > LOAD
CALIBRATE > CALIBRATE > ISOLN
CALIBRATE > CALIBRATE > THRU
CALIBRATE > CALIBRATE > DONE > SAVE